Intermolecular Forces

Answer key with worked solutions

A. Understanding

1. Intermolecular forces.

> (a) An attraction **between** molecules (dispersion, dipole–dipole or hydrogen bonding). (b) Boiling only separates the molecules from each other by overcoming the **intermolecular** forces; the covalent O–H bonds **inside** each molecule are much stronger (about 463 kJ/mol, against 40.7 kJ/mol to vaporize water) and are not broken. Steam is still H_2O.

B. Identifying forces

2. Forces.

> (a) **dispersion** only (b) **dispersion + dipole–dipole** (polar C–Cl) (c) **dispersion + dipole–dipole + hydrogen bonding** (O–H) (d) **dispersion** only (tetrahedral, the C–Cl dipoles cancel) (e) **dispersion + dipole–dipole + hydrogen bonding** (H–F) (f) **dispersion + dipole–dipole** (bent, polar; S is not N, O or F)

C. Boiling points

3. Halogens.

> All are non-polar, so only **dispersion forces** act. Going down the group the molecules have **more electrons** and are larger, so the dispersion forces get stronger and the boiling point rises. At 25 °C: F_2 and Cl_2 are **gases**, Br_2 is a **liquid** (boils at 59 °C), I_2 is a **solid**.

4. Pentane isomers.

> Both have the same number of electrons, but pentane is a **long, unbranched** chain with a large **surface area of contact** between molecules, so its dispersion forces are stronger. Compact, nearly spherical 2,2-dimethylpropane touches its neighbours over a small area, so its dispersion forces are weaker and it boils lower.

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5. Pairs.

> (a) **Ethanol** (78 °C): it has an O–H group and forms hydrogen bonds; methoxymethane (−24 °C) has no O–H. (b) **HBr**: both are polar, but HBr has more electrons, so stronger dispersion forces. (c) NH_3: it forms hydrogen bonds (H on N); PH_3 cannot, even though it is heavier.

6. Ice.

> In ice each water molecule forms **four hydrogen bonds** in a regular, **open hexagonal network** with a lot of empty space. When ice melts, some hydrogen bonds break and the molecules pack **closer together**, so liquid water is **denser** than ice and ice floats.

D. Solubility

7. Solvents.

> (a) **Hexane**: I_2 is non-polar. (b) **Water**: KCl is ionic and is held in solution by ion–dipole attractions. (c) **Water** (it also mixes with hexane to some extent): ethanol forms hydrogen bonds with water. (d) **Hexane**: octane is a non-polar hydrocarbon. Like dissolves like.

E. Calculations

8. Energy to vaporize.

> (a) $n = \dfrac{100.\ \text{g}}{46.07\ \text{g/mol}} = 2.171\ \text{mol}$; $q = 2.171\ \text{mol} \times 38.6\ \text{kJ/mol} = \mathbf{83.8\ kJ}$
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> (b) $n = \dfrac{100.\ \text{g}}{18.02\ \text{g/mol}} = 5.549\ \text{mol}$; $q = 5.549\ \text{mol} \times 40.7\ \text{kJ/mol} = \mathbf{226\ kJ}$
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> (c) The molar values are similar, but 100. g of water contains **far more molecules** (water's molar mass is much smaller), and each one forms **up to four** hydrogen bonds, all of which must be overcome.

9. Per gram.

> Methane: $\dfrac{8.19\ \text{kJ/mol}}{16.04\ \text{g/mol}} = 0.511\ \text{kJ/g}$; water: $\dfrac{40.7\ \text{kJ/mol}}{18.02\ \text{g/mol}} = 2.26\ \text{kJ/g}$
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> Ratio $= \dfrac{2.26\ \text{kJ/g}}{0.511\ \text{kJ/g}} = \mathbf{4.42}$. Methane has only weak **dispersion** forces; water has strong **hydrogen bonds**, so much more energy is needed to separate its molecules, even though the molecules are of similar size.

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